# LAB 03 | Chaotic Text Generator

**Problem and solution.** Generate sentences without reference outputs, compare structural validity, plausibility, novelty and lexical diversity, and decode POS tags from input text. We train count-based first-order HMMs with coarse or role-split hidden tags, plus a trigram baseline. Sampling is controlled by temperature; Viterbi uses smoothed log-probabilities. Data is split by document: 960 synthetic training sentences and 240 test sentences. Metrics use 500 generated sentences per model, seed 42.

## *1. Generation at temperature T = 1.0*

| Model | Valid pattern | Plausible S–V | Novel | Distinct-2 |
|---|---|---|---|---|
| Plain HMM | 16.2% | 14.8% | 99.8% | 0.198 |
| Role-split HMM | 100.0% | 60.6% | 99.4% | 0.148 |
| Trigram | 57.0% | 76.0% | 91.2% | 0.105 |

The plain HMM only sees the preceding coarse POS tag: NOUN can stand for a subject, object or prepositional object, so transitions often form impossible sequences. Splitting DET/NOUN/ADJ into S, O and P roles preserves position in the sentence and gives 100% structural validity on this tiny grammar. The trigram captures local lexical context, improving subject–verb plausibility but sometimes copying training text.

## *2. Temperature selection and failures*

**Writing toy:** T = 1.5 with the role-split HMM: more lexical variety (distinct-2 = 0.161 vs 0.148 at T = 1.0) while keeping 100% valid tag patterns. **Autocomplete:** T = 0.5 with the trigram: more predictable continuations and better structural validity (63.6% vs 44.2% at T = 2.5), though copied text must be flagged.
**Meaning failure:** “the foxes lose by the school .” has valid POS structure but a subject–verb pairing absent from training; role splitting does not model semantics. **Memorisation failure:** “the girls leave the park .” is copied verbatim from training by the low-temperature trigram. The plain HMM also shows structural failures because it merges distinct noun roles.

## *3. Synthetic versus real text*

| Dataset / model | Token acc. | Known | Unknown | Whole sentence |
|---|---|---|---|---|
| Synthetic / plain HMM | 100.0% | — | — | 100.0% |
| UD EWT / most-frequent tag | 86.2% | 90.8% | 28.6% | 31.9% |
| UD EWT / HMM+Viterbi | 88.0% | 92.9% | 28.2% | 40.4% |

The UD EWT figures come from a fresh run of `python -m src.evaluate_real` (12,544 training / 2,077 test sentences; results/ewt_results.json). Real English has 17 POS tags, a much larger vocabulary, 10,359 distinct training tag sequences, and unknown words make up 7.5% of test tokens. The synthetic grammar has only 16 observed patterns and unambiguous context. The HMM gains only 1.8 points over the most-frequent-tag baseline on real tokens; unseen-word emissions and ambiguity (e.g. PROPN→NOUN, 281 errors) cause the rest. So 100% synthetic tagging and valid-pattern scores do not generalise to real English.

## *4. Implementation and limitations*

The Streamlit test bench supports model choice, temperature, seed, 10–500 samples, per-sentence tags and validity, four aggregate metrics, CSV export, a smoothed Viterbi tagger with an OOV warning, and a model/temperature comparison (screenshots on page 2). The generator can produce nonsense or reproduce training sentences; scores are relative to a synthetic corpus, not a measure of real language quality. Reproduction: pip install -r requirements.txt; python -m src.evaluate; python -m src.evaluate_real; streamlit run app.py. Dataset: Universal Dependencies English EWT, CC BY-SA 4.0. Public deployment URL: *to be added by the student after deploying on Streamlit Community Cloud.*

## *Application screenshots (local run)*

Figure 1 – Generation: Plain HMM, T = 1.0, seed 42, 100 sentences.

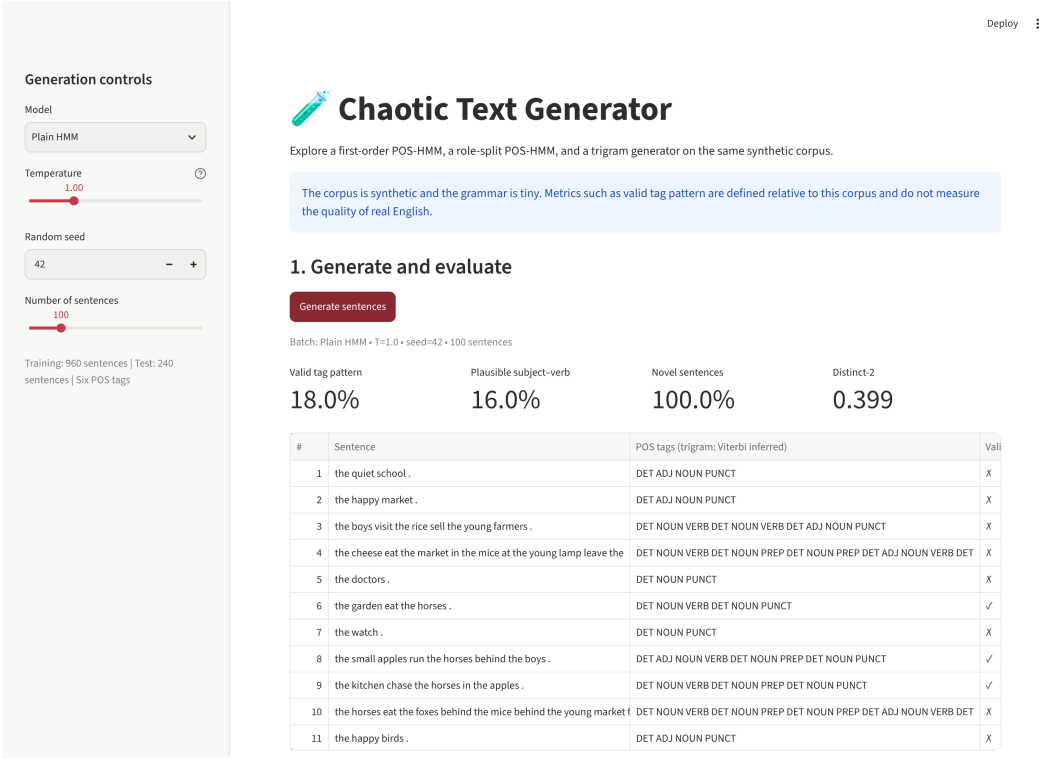



Figure 2 – Viterbi tagger with out-of-vocabulary warning (“zebras”).

